

	Company Name: RAMA CONSTRUCTION PLC	Document No. RAMA-CRD-009	
Title:	Solid and Liquid Waste Management Criteria Document	Revision No: 02	Page 1 of 12

ISSUE HISTORY			
Issue	Description of Change	Originator	Effective Date
3	- Initial Release - Change for update Revised by: DCIC May 26, 2015 Revised as per the Comment on Surveillance Audit 02 and 2015 standard & on March 27, 2017 Revised to upgrade from 18001:2007 to 45001:2018 on 15/02/2020	Dr. Seife-Selassie Ayalew	01/12/2010

REFERENCE DOCUMENTS	
DOCUMENT NUMBER	DOCUMENT TITLE
ISO 14001:2015, Clause No. 8.1	Operational Planning & Control
ISO 9001:2015, Clause 7.5	Documented Information
ISO 9001:2015, Clause 7.1	Resource
ISO 45001:2018, Clause 7.1	Resource

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1. PURPOSE & SCOPE:

PURPOSE

- ◆ To define the responsibility of each function with respect to solid and liquid waste management
- ◆ To includes procedures and practices to prevent discharge of pollutants to the storm drain system or to watercourses or to the soil during collection and disposal of solid and liquid wastes.
- ◆ To indicate the flow of solid and liquid waste management clearly.

SCOPE

This document helps as a guideline to manage all solid and liquid wastes from different activities of the company both at Head Quarters as well as Projects level.

2. CRITERION

2.1 Solid Wastes

S.N.	Types of Wastes	Source of waste
1	Metal Scrap	• Construction site • Garage and workshop
2	Wood Scrap	• Construction site • Workshops
3	Plastic scrap	• Construction site
4	Broken Hollow Concrete Block (HCB)	• Construction site
5	Broken Glasses	• Construction site
6	Used Battery	• Garage
7	Used Tyre	• Garage
8	Paper, Toner and other related solid wastes	• All functions
9	Packing Materials	• Construction site • Store

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2.1.1 Solid Waste Management Flow Chart

Chart No. 01

1. Type of solid waste : **Metal Scrap**
2. Area of waste generation **Construction Site, Garage and Workshop**
3. Management flow chart

S.N.	Activities	Area/Site	Responsibility	Frequency	Documentation
1	Collection	Project site Garage & Workshop	General Foreman Garage/ Workshop Head	Monthly Every day	Metal scrap log sheet
2	Transportation	Project store Central store	General Foreman/ Garage Head	Following each collection	
3	Storage	Project store HQ store	Project Store Head HQ Store Keeper	After being transported Every day	Material return voucher Bin card
4	Disposition 4.1 Sales	Auctioned	Auction committee	At demand	• Auction committee minutes • Payment receipt

Chart No. 02

1. Type of solid waste : **Wood Scrap**
2. Area of waste generation **Construction Site**
3. Management flow chart

S.N.	Activities	Area/Site	Responsibility	Frequency	Documentation
1	Collection	Project site	General Foreman	Upon completion of project or when the wood can no more give service	Wood scrap log sheet
2	Transportation	Project store	General Foreman	Following each collection	Material return voucher
3	Storage	Project store allocated area	Project Store Keeper		
4	Disposition 4.1 Sales	Auctioned Sold to employees Given to employees	Auction committee	Upon completion of project or when the need arises	• Auction committee minutes • Payment receipt • Letter instructing the donation

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Chart No. 03

1. Type of solid waste : **Plastic Scrap**
2. Area of waste generation **Construction Site, camp and Store**
3. Management flow chart

S.N.	Activities	Area/Site	Responsibility	Frequency	Documentation
1	Collection	Project site Store	General Foreman Project Camp Admin Store Head	At demand	Plastic scrap log sheet
2	Transportation	Project store	General Foreman Project camp Admin	Following each collection	Material return voucher (damage)
3	Storage	Project store	Project Store Keeper		
	Disposition 4.1 Sales	Auctioned	Auction committee	When the volume of such scrap reaches a point where it could be sold	• Auction committee minutes • Payment receipt
	4.2 Recycling	Recycling Plant	Store Head	Once every year	Recyclable materials delivery note
	4.3 Burning	In the incinerator & Municipality land fill	Janitors	At demand	

Chart No. 04

1. Type of solid waste **Broken Hollow Concrete Blocks (HCB)**
2. Area of waste generation **Construction Site**
3. Management flow chart

S.N.	Activities	Area/Site	Responsibility	Frequency	Documentation
1	Collection	Project site	General Foreman	Monthly	
2	Transportation	Crushing machine	General Foreman	After each collection	
3	Storage	Near crushing machine at allocated area	HCB Production foreman		
4	Disposition 4.1 Recycling	Near crushing machine at allocated area	HCB Production foreman	At demand	Crushed materials (HCB) Log sheet

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Chart No. 05

1. Type of solid waste **Broken Glasses**
2. Area of waste generation **Construction Site**
3. Management flow chart

S.N.	Activities	Area/Site	Responsibility	Frequency	Documentation
1	Collection	Project site	General Foreman	At demand	Damaged Materials Registration and Reporting Form
2	Transportation	Project Store	General Foreman	At demand	
3	Storage	In an area designated within or next to the store	Store Head	At demand	
4	Disposition				
	4.1 Reuse	In a workshop	Glazing person	At demand	Recycled glasses log sheet
	4.2 Disposal	In a safe area specially made for this purpose	Glazing Person	At demand	

Chart No. 06

1. Type of solid waste **Used Battery**
2. Area of waste generation **Garage**
3. Management flow chart

S.N.	Activities	Area/Site	Responsibility	Frequency	Documentation
1	Collection	Garage	Equipment Admin	At demand	Returned Battery Log Sheet
2	Transportation	Store	Equipment Admin & Store Head	At demand	
3	Storage	Store	Battery Administrator	At demand	Material Return Voucher
4	Disposition				
	4.1 Sales	Auctioned	Auction committee	Every Year	• Auction committee minutes • Payment receipt

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Chart No. 07

4. Type of solid waste **Used Tyre**
5. Area of waste generation **Garage**
6. Management flow chart

S.N.	Activities	Area/Site	Responsibility	Frequency	Documentation
1	Collection	Garage	Equipment Admin	At demand	Returned Tyre Log Sheet
2	Transportation	Store	Equipment Admin & Store Head	At demand	Transfer Voucher
3	Storage	Store	Tyre Administrator	At demand	Material Return Voucher
4	Disposition				
	4.1 Retreading/Recapping	Tyre Retreading or Recapping Plant	Equipment Admin	At Demand	- Retreading or recapping - Delivery Note - Out Side Repair order
	4.2 Sales	Auctioned	Auction committee	Every Year	- Auction committee minutes - Payment receipt

Chart No. 08

1. Type of solid waste **Paper, Toner and other related solid wastes**
2. Area of waste generation **All functions**
3. Management flow chart

S.N.	Activities	Area/Site	Responsibility	Frequency	Documentation
1	Collection	All functions offices	- Store Head - Janitors	At Demand	
2	Transportation	To the incinerator, container & Tractor	- General Service - Janitors	At Demand	
3	Storage	To the incinerator, container & Tractor	- General Service - Janitors	At Demand	
4	Disposition				
	4.1 Selling	In the container	Auction Committee	Once a year	- Auction committee minutes - Payment receipt
	4.2 Burning	In the incinerator & Municipality land fill	Janitors	At demand	
	4.3 Recycling	Toner re-filler company	General Service	Once every six months	- Re-filler Company Goods Delivery Note - Goods Receiving Voucher

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Chart No. 09

1. Type of solid waste **Packing Materials**
2. Area of waste generation **Construction Site, Store**
3. Management flow chart

S.N.	Activities	Area/Site	Responsibility	Frequency	Documentation
1	Collection	Construction Site Store	• General Foreman • Store Head	At Demand	
2	Transportation	To the incinerator, solid waste container, Store	• Porters • Janitors	At Demand	
3	Storage	Incinerator, container, Store	• General Service • Janitors • Store Head	At Demand	
4	Disposition				
	4.1 Selling	In the container	Auction Committee	Once a year	• Auction committee minutes • Payment receipt
	4.2 Burning	In the incinerator & Municipality land fill	Janitors	At demand	

2.2 Liquid Wastes

S.N.	Types of Wastes	Source of waste
1	Left over paints	• Construction site
2	Grease-free and oil-free waste water and rinse water	• Construction site • Work shop
3	Used Oils	• Garages • Work Shops

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2.2.1 Liquid (Fluid) Waste Management Flow Chart

Chart No. 01

1. Type of Liquid waste : **Used Oils**
2. Area of waste generation **Construction Site, Garage and Workshop**
3. Management flow chart **Liquid Waste (waste oils)**
4. Type of solid waste Area of waste generation **Construction Site, maintenance of site equipment (Garages) grease or Oils traps**
5. Management flow chart

S.N.	Activities	Area/Site	Responsibility	Frequency	Documentation
1	Collection	Construction Site Store, Main Store, cleaning and maintenance of site equipment (Garage)/ grease/Oils traps	• General Foreman • Store Head • Maintenance Head • Lubricant Administrator	At Demand	
2	Transportation	To the incinerator, liquid waste container, Store	• Store Heads • Garage Heads • Lub Admin	At Demand	
3	Storage	To used oils collection Drums, grease or Oils traps	• General Service • Janitors • Store Head • Grease boy	At Demand	
4	Disposition				
	4.1 Reuse 4.2 Selling	For formwork In the container	• Construction Forman • Auction Committee	At demand	• Transfer Voucher • Auction committee minutes • Payment receipt

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Chart No. 02

1. Type of Liquid waste : **Expired/ Leftover Paints**
2. Area of waste generation **Construction Site and Workshop**
3. Management flow chart **Liquid Waste including (paints and solvents)**
4. Type of Liquid waste Area of waste generation **Construction Site, Store, cleaning and Workshops**
5. Management flow chart

S.N.	Activities	Area/Site	Responsibility	Frequency	Documentation
1	Collection	Construction Site Store, Main Store, cleaning and workshops	• Gen Foreman • Store Head • Shop Head • Lub Admin	At Demand	
2	Transportation	To the incinerator, liquid waste container, Store	• Store Heads • Shop Heads • Lub Admin	At Demand	
3	Storage	To collection Drums, with its own container	• General Service • Janitors • Store Head • Grease boy	At Demand	
4	Disposition				
	4.1 Reuse 4.2 Selling	For formwork In the container	• Construction Forman • Auction Committee	At demand	• Transfer Voucher • Auction committee minutes • Payment receipt

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Chart No. 03

1. Type of Liquid waste : **Expired/ Leftover Solvents**
2. Area of waste generation **Construction Site and Workshop**
3. Management flow chart **Liquid Waste including (waste oil, lubricants, paints and solvents)**
4. Type of Liquid waste Area of waste generation **Construction Site, Store, cleaning and Workshops**
5. Management flow chart

S.N.	Activities	Area/Site	Responsibility	Frequency	Documentation
1	Collection	Construction Site Store, Main Store, cleaning and workshops	• Gen Foreman • Store Head • workshop Head • Lub Admin	At Demand	
2	Transportation	To the incinerator, liquid waste container, Store	• General forman • Store Heads • Garage Heads • Lub Admin	At Demand	
3	Storage	To used oils collection Drums, grease or Oils traps	• General Service • Janitors • Store Head • Grease boy	At Demand	
4	Disposition				
	4.1 Reuse 4.2 Selling	For formwork In the container	• Construction Forman • Auction Committee	At demand	• Transfer Voucher • Auction committee minutes • Payment receipt

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2.3. NATURE OF WASTE & METHOD OF DISPOSITION:

2.3.1 Solid & Liquid Wastes

S.N.	Types of Wastes	Description	Method of disposal
1	Metal Scrap	Used or wasted metal & spare parts	Sales
2	Wood Scrap	Used or wasted wood	Sales Donation to employees
3	Plastic Scrap	Used or wasted, due to damage, plastic materials	Sales Recycling Burning
4	Broken Hollow Concrete Blocks (HCB)	Damaged or broken HCB	Recycling through crushing
5	Broken Glasses	Broken glasses in use or during transportation	Reuse Disposal
6	Used Battery	Car batteries service life of which is over	Sales
7	Used Tyre	Tyres which have served their life	Retreading/Recapping Sales
8	Paper, Toner and other related solid wastes	Unwanted paper	Burning
		Refillable toner	Recycling
		Resalable items	Sales
9	Packing Materials	Different kind of packing materials	Selling Burning
10	Left over paints	Liquid Waste including (paints and solvents)	Reuse Selling
11	Grease-free and oil-free waste water and rinse water	Liquid Waste (waste oils)	Reuse Selling
12	Used Oils	Liquid Waste including (waste oil, lubricants, paints and solvents)	Reuse Selling

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2.4. RELATED DOCUMENT

DOCUMENT TITLE

Solid & Liquid Waste Registration Logging sheet
 Solid & Liquid Waste Disposal Form
 Auction Committee Minutes
 Returned Used Tyre Log Sheet
 Returned Car Battery Log Sheet
 Damaged Glass Reuse Form
 Out Side Repair Order form
 Receiving Voucher
 Transfer Voucher

DOCUMENT NO.

RAMA-OF-037
 RAMA-OF-038
 RAMA-OF-043
 RAMA-OF-040
 RAMA-OF-041
 RAMA-OF-042
 RAMA-OF-5232
 RAMA-OF-301
 RAMA-OF-306

3. DISTRIBUTION

- ◆ Managing Director
- ◆ Chief Operations Officer
- ◆ Plant and Equipment Operations Officer
- ◆ General Manager
- ◆ All Departments
- ◆ All Projects

Approval Name: <i>Ferew Tedla Managing Director</i>	Signature: 	Date: 15/02/2020
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